

Disproof of conjecture 00000000462

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Verdict: FALSE

We disprove conjecture 00000000462 as stated. Let $\tau(n)$ denote the number of spanning trees of the circulant graph $C_n(1, 2)$, the *square of the n -cycle*: the graph on $\mathbb{Z}/n$ joining i to $i \pm 1$ and to $i \pm 2$. We compute τ exactly, and show:

1. τ satisfies the six-term linear recurrence

$$\tau(n+6) = 4\tau(n+5) - 10\tau(n+3) + 4\tau(n+1) - \tau(n),$$

whose characteristic polynomial is

$$p(x) = x^6 - 4x^5 + 10x^3 - 4x + 1 = ((x+1)(x^2 - 3x + 1))^2.$$

Its largest real root is

$$\frac{3 + \sqrt{5}}{2} = \varphi^2 = 2.6180339887\dots, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

2. The conjectured value

$$\alpha^2 = (2 + \sqrt{3})^2 = 7 + 4\sqrt{3} = 13.9282032302\dots$$

is *not* a root of p : an exact computation in $\mathbb{Z}[\sqrt{3}]$ gives

$$p(7 + 4\sqrt{3}) = 2615536 + 1510080\sqrt{3} \neq 0.$$

Consequently α^2 cannot be a characteristic root of the recurrence satisfied by τ , and it is also not the exponential growth rate, which is φ^2 . The conjecture fails under either reading of its wording; see Section 5.

1 The conjecture

Quoted verbatim from `conjectures/00000000462.md`:

English. Definition: $\tau(\text{circulant } C_n(1, 2))$. Conjecture: $\tau(C_n(1, 2))$ satisfies a linear recurrence whose largest real characteristic root tends to α^2 , where $\alpha = 2 + \sqrt{3}$ (directly verifiable against the matrix-tree theorem, then stateable in closed form).

The Chinese statement filed with the conjecture is reproduced verbatim in `README.md` together with this English text; it is not typeset here so that `main.tex` has no CJK font requirement and compiles with a plain `tectonic main.tex`. The two formulations are translations of one another, and both are refuted below.

Definition 1 (The graph). For $n \geq 5$ let $C_n(1, 2)$ be the circulant graph on $\mathbb{Z}/n\mathbb{Z}$ in which i and j are adjacent whenever $j - i \equiv \pm 1$ or $j - i \equiv \pm 2 \pmod{n}$. Let $\tau(n) = \tau(C_n(1, 2))$ be its number of spanning trees.

Remark 2. For $n = 3$ and $n = 4$ the connection set $\{\pm 1, \pm 2\}$ collapses ($\pm 2 = \mp 1$ for $n = 3$, and $\pm 2 = \{2\}$ with $\pm 1 = \{1, 3\}$ for $n = 4$), and the graphs are the complete graphs K_3 and K_4 with $\tau(3) = 3$ and $\tau(4) = 16$. The closed form below is the one for the general case $n \geq 5$, which is the case the conjecture is about. Every statement we prove is checked numerically for $5 \leq n \leq 60$.

2 The exact spanning-tree count

Theorem 3 (Closed form). *For every $n \geq 5$,*

$$\tau(n) = \frac{n(L(2n) + 2(-1)^{n-1})}{5},$$

where L is the Lucas sequence $L(0) = 2$, $L(1) = 1$, $L(k+2) = L(k+1) + L(k)$. In particular $\tau(n)$ is a positive integer and $5 \mid L(2n) + 2(-1)^{n-1}$.

Proof. By the matrix-tree theorem (Kirchhoff), for a connected graph on n vertices with Laplacian eigenvalues $\lambda_0 = 0, \lambda_1, \dots, \lambda_{n-1}$,

$$\tau = \frac{1}{n} \prod_{k=1}^{n-1} \lambda_k.$$

For a circulant graph with connection set S , the Laplacian eigenvalues are $\lambda_k = \sum_{s \in S} (1 - \omega^{ks})$ with $\omega = e^{2\pi i/n}$. Here $S = \{\pm 1, \pm 2\}$, so with $z = \omega^k$,

$$\lambda_k = 4 - z - z^{-1} - z^2 - z^{-2} = -\frac{(z-1)^2(z^2 + 3z + 1)}{z^2}, \quad (1)$$

the factorisation following from $z^4 + z^3 - 4z^2 + z + 1 = (z-1)^2(z^2 + 3z + 1)$.

Set $z_k = \omega^k$ for $k = 1, \dots, n-1$; these are the nontrivial n -th roots of unity, and $\prod_{k=1}^{n-1} z_k^2 = \omega^{2 \cdot n(n-1)/2} = 1$. Also

$$\prod_{k=1}^{n-1} (z_k - 1) = (-1)^{n-1} \prod_{k=1}^{n-1} (1 - \omega^k) = (-1)^{n-1} n,$$

because $\prod_{k=1}^{n-1} (x - \omega^k) = (x^n - 1)/(x - 1)$ evaluated at $x = 1$, whence

$$\prod_{k=1}^{n-1} (z_k - 1)^2 = n^2. \quad (2)$$

Let r_1, r_2 be the roots of $z^2 + 3z + 1$, so $r_1 + r_2 = -3$ and $r_1 r_2 = 1$. Since $\prod_{k=1}^{n-1} (x - z_k) = (x^n - 1)/(x - 1)$ for $x \notin \{z_k\}$,

$$\prod_{k=1}^{n-1} (z_k - r) = (-1)^{n-1} \frac{r^n - 1}{r - 1},$$

hence

$$\prod_{k=1}^{n-1} (z_k^2 + 3z_k + 1) = \prod_{k=1}^{n-1} (z_k - r_1)(z_k - r_2) = \frac{(r_1^n - 1)(r_2^n - 1)}{(r_1 - 1)(r_2 - 1)} = \frac{(r_1^n - 1)(r_2^n - 1)}{5}, \quad (3)$$

because $(r_1 - 1)(r_2 - 1) = r_1 r_2 - (r_1 + r_2) + 1 = 1 + 3 + 1 = 5$. Combining (1), (2) and (3),

$$\tau(n) = \frac{1}{n} \prod_{k=1}^{n-1} \lambda_k = \frac{1}{n} \cdot (-1)^{n-1} \cdot n^2 \cdot \frac{(r_1^n - 1)(r_2^n - 1)}{5} = \frac{(-1)^{n-1} n (r_1^n - 1)(r_2^n - 1)}{5}.$$

Now $(r_1^n - 1)(r_2^n - 1) = (r_1 r_2)^n - r_1^n - r_2^n + 1 = 2 - (r_1^n + r_2^n)$. Writing $s_1 = \frac{3-\sqrt{5}}{2}$ and $s_2 = \frac{3+\sqrt{5}}{2}$ we have $r_1 = -s_1$, $r_2 = -s_2$, so $r_1^n + r_2^n = (-1)^n (s_1^n + s_2^n)$. Since s_1, s_2 are the roots of $x^2 - 3x + 1$ with $s_1^0 + s_2^0 = 2$ and $s_1 + s_2 = 3$, the sequence $s_1^n + s_2^n$ is $L(2n)$: indeed $L(2n)$ satisfies the same recurrence $x^2 - 3x + 1$ with initial values $L(0) = 2$, $L(2) = 3$. Therefore

$$(r_1^n - 1)(r_2^n - 1) = 2 - (-1)^n L(2n),$$

and

$$\tau(n) = \frac{(-1)^{n-1} n (2 - (-1)^n L(2n))}{5} = \frac{n(L(2n) + 2(-1)^{n-1})}{5},$$

since $-(-1)^{n-1}(-1)^n = +1$. This is the asserted form. $\square$

The integrality statement in Theorem 3 is an instance of a small congruence: $L(2n) \equiv 2 \pmod{5}$ for even n and $L(2n) \equiv 3 \pmod{5}$ for odd n , so $L(2n) + 2(-1)^{n-1} \equiv 2 - 2 \equiv 0 \pmod{5}$ (even n) and $\equiv 3 + 2 \equiv 0 \pmod{5}$ (odd n) modulo 5.

3 The linear recurrence and its characteristic polynomial

Write $u(n) = L(2n) + 2(-1)^{n-1}$, so that $\tau(n) = n u(n)/5$ by Theorem 3.

Lemma 4 (The recurrence for u). *For every $n \geq 0$,*

$$u(n+3) = 2u(n+2) + 2u(n+1) - u(n).$$

Equivalently, u satisfies the linear recurrence with characteristic polynomial

$$q(x) = x^3 - 2x^2 - 2x + 1 = (x+1)(x^2 - 3x + 1).$$

Proof. The even-index Lucas numbers satisfy $L(2n+4) = 3L(2n+2) - L(2n)$; this is the standard Lucas identity $L(m+4) + L(m) = 3L(m+2)$ read at $m = 2n$, and it is the recurrence with characteristic polynomial $x^2 - 3x + 1$. The sequence $(-1)^{n-1}$ satisfies $x + 1$. Summing the two recurrences in operator form gives

$$(E^2 - 3E + 1)(E + 1)u = 0,$$

i.e. $(E^3 - 2E^2 - 2E + 1)u = 0$, which is the stated three-term recurrence. The factorisation $x^3 - 2x^2 - 2x + 1 = (x+1)(x^2 - 3x + 1)$ is verified by expansion: $(x+1)(x^2 - 3x + 1) = x^3 - 3x^2 + x + x^2 - 3x + 1 = x^3 - 2x^2 - 2x + 1$. $\square$

Lemma 5 (The recurrence for τ). *For every $n \geq 0$ the sequence τ satisfies*

$$\tau(n+6) = 4\tau(n+5) - 10\tau(n+3) + 4\tau(n+1) - \tau(n),$$

so its characteristic polynomial divides

$$p(x) = q(x)^2 = x^6 - 4x^5 + 10x^3 - 4x + 1 = ((x+1)(x^2 - 3x + 1))^2.$$

Proof. Multiplying a sequence satisfying $q(E)w = 0$ by n doubles the multiplicity of every root: if $q(E)w = 0$ then $q(E)^2(nw) = 0$, because $q(E)(nw(n))$ is a fixed finite linear combination of the shifts of w , which $q(E)$ annihilates. Applying this to $w = u$ and using $\tau(n) = nu(n)/5$ gives $p(E)\tau = 0$ with $p = q^2$.

For $q(x) = x^3 - 2x^2 - 2x + 1$ we compute

$$q(x)^2 = (x^3 - 2x^2 - 2x + 1)^2 = x^6 - 4x^5 + 10x^3 - 4x + 1.$$

Read as an operator identity $E^6 - 4E^5 + 10E^3 - 4E + 1 = 0$, this is exactly the displayed recurrence. (The computation was also verified symbolically and numerically; see `reproduce.py`.) $\square$

Theorem 6 (Roots). *The roots of p are -1 (twice) and*

$$s_2 = \frac{3 + \sqrt{5}}{2} = \varphi^2 \approx 2.6180339887, \quad s_1 = \frac{3 - \sqrt{5}}{2} \approx 0.3819660113,$$

each twice (the roots of $x^2 - 3x + 1$). The largest real root is $s_2 = \varphi^2$ and it is simple as a root of p in the sense that its multiplicity as a root of the minimal annihilator of τ is at least the multiplicity forced by the factor n ; the exponential growth rate of τ is φ^2 .

Proof. Immediate from $p = ((x+1)(x^2 - 3x + 1))^2$ and the quadratic formula. The growth statement follows from Theorem 3: $\tau(n) = nu(n)/5$ and $u(n) = s_1^n + s_2^n + 2(-1)^{n-1}$ with $|s_1| < 1 < s_2$, so $\tau(n) = (s_2^n(1 + o(1))) \cdot n/5$ and $\lim_{n \rightarrow \infty} \tau(n)^{1/n} = s_2 = \varphi^2$. $\square$

4 The conjectured root is not a characteristic root

Theorem 7. $\alpha^2 = (2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$ is not a root of $p(x) = x^6 - 4x^5 + 10x^3 - 4x + 1$.

Proof. Because $p(x) = (x+1)^2(x^2 - 3x + 1)^2$, a number is a root of p iff it equals -1 or is a root of $x^2 - 3x + 1$. Work in the ring $\mathbb{Z}[\sqrt{3}] \cong \mathbb{Z}[t]/(t^2 - 3)$, embedded in $\mathbb{R}$ by $t \mapsto \sqrt{3}$.

With $\alpha = 2 + \sqrt{3}$ we get $\alpha^2 = 7 + 4\sqrt{3}$ and $\alpha^4 = (\alpha^2)^2 = 97 + 56\sqrt{3}$ in $\mathbb{Z}[\sqrt{3}]$. Hence

$$\alpha^4 - 3\alpha^2 + 1 = (97 + 56\sqrt{3}) - 3(7 + 4\sqrt{3}) + 1 = 77 + 44\sqrt{3}.$$

Since $77 > 0$ and $44\sqrt{3} > 0$, this number is nonzero. Also $\alpha^2 = 7 + 4\sqrt{3} > 0 > -1$. Therefore

$$p(\alpha^2) = (\alpha^2 + 1)^2(\alpha^4 - 3\alpha^2 + 1)^2 = (8 + 4\sqrt{3})^2(77 + 44\sqrt{3})^2 > 0,$$

and expanding, $p(\alpha^2) = 2615536 + 1510080\sqrt{3} \neq 0$. $\square$

Corollary 8 (Refutation). *The sequence $\tau(C_n(1, 2))$ satisfies a linear recurrence with characteristic polynomial p , and the conjectured value $\alpha^2 = 7 + 4\sqrt{3}$ is not a root of p . Since the minimal annihilating polynomial of a linear recurrence divides every annihilating polynomial, every characteristic root of τ is a root of p ; hence α^2 is not a characteristic root of any linear recurrence satisfied by τ . The conjecture is false as stated.*

Proof. Lemma 5 exhibits the annihilating polynomial p of τ ; Theorem 7 shows α^2 is not a root of p . If q is the minimal annihilating polynomial of τ then $q \mid p$, so every root of q is a root of p ; since $p(\alpha^2) \neq 0$ we get $q(\alpha^2) \neq 0$. $\square$

5 The two readings of the conjecture

The filed wording says that τ “satisfies a linear recurrence whose largest real characteristic root tends to α^2 ”. The word “tends” is not standard for characteristic roots of a constant-coefficient recurrence; we dispose of both natural readings.

1. **“Characteristic root” = root of the characteristic polynomial.** For a linear recurrence $a_{n+k} = c_{k-1}a_{n+k-1} + \dots + c_0a_n$ the characteristic roots are the roots of $x^k - c_{k-1}x^{k-1} - \dots - c_0$, i.e. the roots of any annihilating polynomial. By Corollary 8 every such root of τ is a root of p , and α^2 is not one. *The conjecture is false.*
2. **“Tends to” = the exponential growth rate is α^2 .** If the claim is that $\lim_{n \rightarrow \infty} \tau(n)^{1/n} = \alpha^2$, then the limit is in fact $\varphi^2 = 2.6180339887\dots$ by Theorem 6, not $\alpha^2 = 13.9282032302\dots$. Indeed $\tau(n) \sim n\varphi^{2n}/5$: for $n = 60$ the ratio $\frac{\tau(60)/60}{\tau(59)/59}$ equals 2.6180339887 to ten decimal places. *The conjecture is false.*

Remark 9 (Why the numbers differ). $\varphi^2 = \frac{3+\sqrt{5}}{2}$ comes from the Lucas recurrence $x^2 - 3x + 1$, which is the recurrence genuinely satisfied by the even-index Lucas numbers underlying $u(n) = L(2n) + 2(-1)^{n-1}$. The filed value $\alpha^2 = 7 + 4\sqrt{3}$ comes from the order-2 algebraic number $2 + \sqrt{3}$; it is not related to $C_n(1, 2)$ in any way we can detect, and $p(\alpha^2) \neq 0$ rules it out as a root of the annihilator. Note also that $\alpha = 2 + \sqrt{3}$ is *not* of the form φ^k for integer k ($\varphi^3 = 2 + \sqrt{5}$), so it cannot arise from the Lucas family either.

6 Numerical verification

The following table collects the first values. The “matrix tree” column is the exact integer determinant of the reduced Laplacian (fraction-free Bareiss elimination); the “closed form” column is $n(L(2n) + 2(-1)^{n-1})/5$.

n	matrix tree $\tau(n)$	closed form	recurrence residual
5	125	125	0
6	384	384	0
7	1183	1183	0
8	3528	3528	0
9	10404	10404	0
10	30250	30250	0
11	87131	87131	0
12	248832	248832	0

The residual column is $\tau(n) - (4\tau(n-1) - 10\tau(n-3) + 4\tau(n-5) - \tau(n-6))$.

The two candidate roots compare as follows.

quantity	value	status
$\varphi^2 = (3 + \sqrt{5})/2$	2.6180339887...	root of p (the actual growth rate)
$\alpha^2 = 7 + 4\sqrt{3}$	13.9282032302...	not a root: $p(\alpha^2) = 2615536 + 1510080\sqrt{3}$

7 Formalisation

The refutation is formalised in the Lean 4 project under `lean4/` (core Lean only — no Mathlib, no `sorry`, no `native_decide`). The project defines the Lucas numbers, the sequence $u(n) = L(2n) + 2(-1)^{n-1}$, proves u 's three-term recurrence, derives the six-term recurrence for $c(n) = nu(n) = 5\tau(n)$, proves c and τ satisfy it, proves $5 \mid u(n)$, verifies the initial values $\tau(5), \dots, \tau(12)$, verifies the factorisation of p by exact coefficient arithmetic, models $\mathbb{Z}[\sqrt{3}]$ as a pair ring, and checks $p(\alpha^2) \neq 0$. The main formal statements are

```
theorem u_rec (n) : u (n+3) = 2*u(n+2) + 2*u(n+1) - u n
theorem tau_rec (n) : tau (n+6) = 4*tau(n+5) - 10*tau(n+3) + 4*tau(n+1) -
tau n
theorem char_square : pmul [1,-2,-2,1] [1,-2,-2,1] = [1,-4,0,10,0,-4,1]
theorem pEval_alpha_sq : pEval (mul alpha alpha) = (2615536, 1510080)
theorem conjecture_00000000462_false :
  (forall n, tau (n+6) = 4*tau(n+5) - 10*tau(n+3) + 4*tau(n+1) - tau n)
  and (forall x, x = mul alpha alpha -> pEval x != zero)
  and mul alpha alpha != scale (-1) one
```

The axiom audit (`lake env lean Check.lean`) reports only `propext` and `Quot.sound` for every theorem, and no `sorryAx`. See `lean4/README.md` for the full statement table and scope note.

8 Reproducibility

```
$ python3 reproduce.py
... (matrix-tree determinant for 5 <= n <= 60, the closed form,
... the recurrence, the factorisation, and the exact check in  $\mathbb{Z}[\sqrt{3}]$ )
PASS: every check verified; conjecture 00000000462 is FALSE.
```

`reproduce.py` uses only the Python 3 standard library and exact integer arithmetic throughout (fraction-free Bareiss elimination for the determinant, integer coefficient lists for polynomials, pairs for $\mathbb{Z}[\sqrt{3}]$ and $\mathbb{Z}[\sqrt{5}]$); it runs in under a second, prints `PASS` and exits 0 exactly when every check holds. The Lean project is built and audited by

```
cd lean4
lake build
lake env lean Check.lean
```

and the document is compiled by

```
tectonic main.tex
(equivalently: tectonic --outdir build main.tex)
```

producing `build/main.pdf`.

9 Status against the submission rules

- **LaTeX source** — present (`main.tex`), a standalone article using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `parskip`, `booktabs`, and `array`; it compiles with `tectonic main.tex`.
- **PDF document** — present at `build/main.pdf`.
- **Lean 4 project** — present under `lean4/`, with `lean-toolchain` (pinned to `leanprover/lean4:v4.33.1`), `lakefile.toml` (library `Main`, project `tlmc462`), `Main.lean` (core Lean only, no `sorry`, no `native_decide`), `Check.lean` (`#print axioms audit`), and `README.md`. The audit reports only `[propext, Quot.sound]` and no `sorryAx`.
- **Auxiliary code** — `reproduce.py`, dependency-free, exits 0 and prints `PASS`.
- **No other files touched** — the pull request adds only the folder `solutions/00000000462/earthking11_sub` and modifies nothing else.